IHLP® Commercial Inductors, High Temperature (155 °C) Series



FEATURES

- High temperature, up to 155 °C
- 10.8 mm x 10.2 mm x 4.0 mm SMD package
- Magnetically shielded construction
- Metal alloy core
- Polarity marking available for EMI sensitive applications (see “EP” package code below for more information)
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

Pb-free
RoHS COMPLIANT
HALOGEN FREE GREEN (5-2008)

LINKS TO ADDITIONAL RESOURCES

Product Page | 3D Models | Models | Calculators

APPLICATIONS

- DC/DC power supplies
- Smart grid and solar
- Telecommunications equipment
- Noise suppression and filtering

STANDARD ELECTRICAL SPECIFICATIONS

PART NUMBER	INDUCTANCE ± 20 % (µH) AT 0 A	DCR 25 °C (mΩ)		HEAT RATING CURRENT DC TYP. (A) (1)	SATURATION CURRENT DC TYP. (A) (2)		SRF TYP. (MHz)
		TYP.	MAX.		20 % DROP	30 % DROP	
IHLP4040DZ**R47M51	0.47	1.55	1.66	35.5	28.5	38.0	72.1
IHLP4040DZ**R68M51	0.68	2.17	2.32	35.0	24.0	32.0	42.5
IHLP4040DZ**1R0M51	1.0	2.87	3.07	23.5	24.0	32.0	37.2
IHLP4040DZ**1R5M51	1.5	4.20	4.50	22.0	17.9	24.2	32
IHLP4040DZ**2R2M51	2.2	8.15	8.76	15.0	12.0	16.2	30.1
IHLP4040DZ**3R3M51	3.3	11	11.81	11.0	12.0	16.2	25.5
IHLP4040DZ**4R7M51	4.7	14.3	15.32	9.8	9.2	12.4	20.1
IHLP4040DZ**5R6M51	5.6	16.5	17.60	9.3	9.0	12.2	16.3
IHLP4040DZ**6R8M51	6.8	20.9	22.36	8.0	9.0	12.2	16.3
IHLP4040DZ**100M51	10	30.9	33.06	6.5	8.5	11.5	11.5
IHLP4040DZ**150M51	15	47	50.29	5.1	7.7	10.4	10.4
IHLP4040DZ**220M51	22	70.5	75.44	4.1	6.4	8.6	8.3
IHLP4040DZ**330M51	33	110	117.7	3.7	4.2	5.7	5.79
IHLP4040DZ**470M51	47	167	178	3.1	4.1	5.5	5.22
IHLP4040DZ**680M51	68	240	252	2.4	3.5	4.7	4.02

Notes

- All test data is referenced to 25 °C ambient
- Test condition: 100 kHz, 0.25 V
- Operating temperature range -55 °C to +155 °C
- The part temperature (ambient + temp. rise) should not exceed 155 °C under worst case operating conditions. Circuit design, component placement, PCB trace size and thickness, airflow and other cooling provisions all affect the part temperature. Part temperature should be verified in the end application
- Rated operating voltage (across inductor) = 75 V

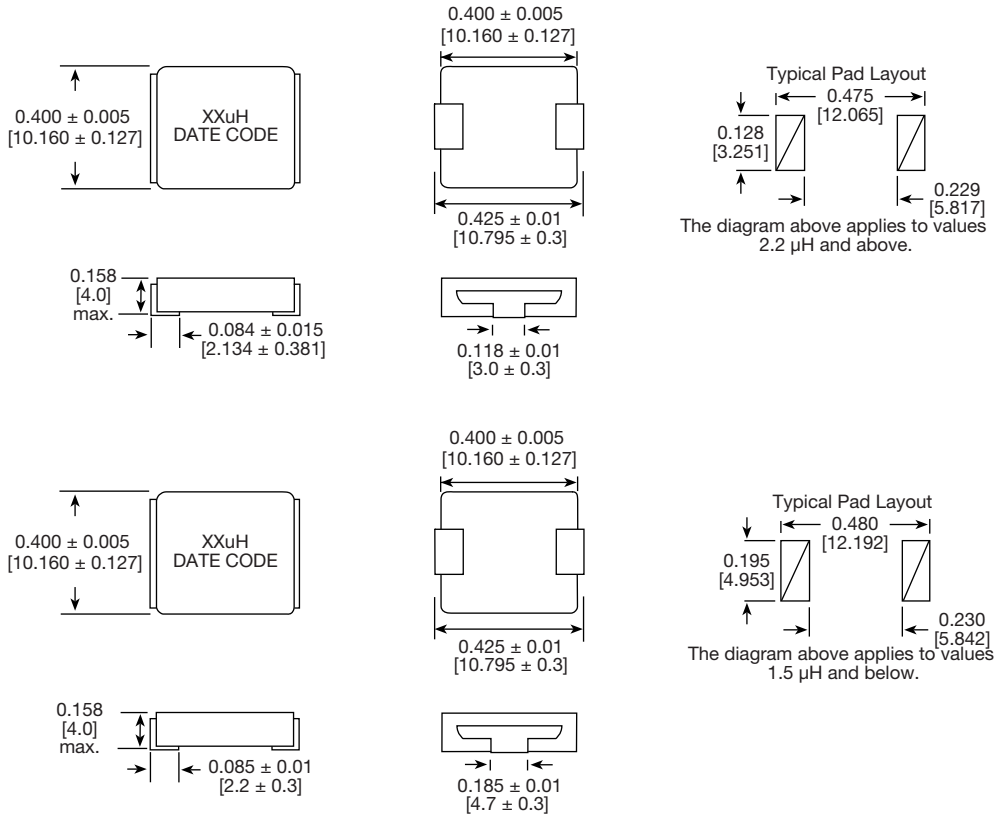
(1) DC current (A) that will cause an approximate ΔT of 40 °C

(2) DC current (A) that will cause L_0 to drop approximately 20 % and 30 %

For technical questions, contact: magnetics@vishay.com

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DIMENSIONS in inches [millimeters]



DESCRIPTION

IHLP-4040DZ-51	4.7 µH	± 20 %	TAPE AND REEL	e3
MODEL	INDUCTANCE VALUE	INDUCTANCE TOLERANCE	PACKAGE CODE	JEDEC® LEAD (Pb)-FREE STANDARD

GLOBAL PART NUMBER

I H L P	4 0 4 0 D Z	E K	4 R 7	M	5 1
PRODUCT FAMILY	SIZE	PACKAGE CODE	INDUCTANCE VALUE	INDUCTANCE TOLERANCE	SERIES
		EK = tape and reel	**4R7** = 2.0 µH	**M** = ± 20 %	

PACKAGE CODE OPTIONS

EK = tape and reel packaging (1000 pieces on 13-inch reel)

EE = tape and reel packaging (900 pieces on 13-inch reel)

ER = tape and reel packaging (500 pieces on 13-inch reel)

EP = tape and reel packaging (1000 pieces on 13-inch reel), includes polarity part marking

PE = tape and reel packaging (900 pieces on 13-inch reel), includes polarity part marking

Notes

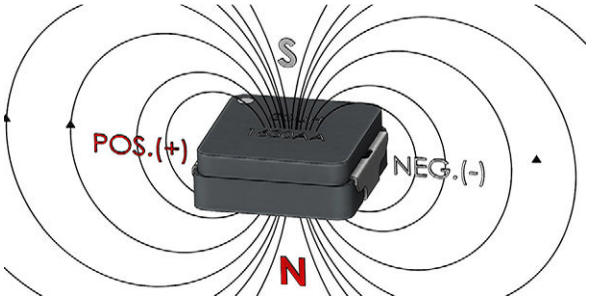
- 1000 piece reels for IHLP4040DZ models contain narrower tape pitch
- For additional packaging details see "Packaging Methods"

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MAGNETIC FIELD

CONFIGURATION OF THE "B" (FLUX) FIELD FOR THE IHLP WITH POLARITY MARKING



When a positive (+) voltage is placed on the terminal marked with the polarity dot and the opposite terminal is negative (-), the resulting current flow will create a magnetic south pole on the top side of the IHLP.

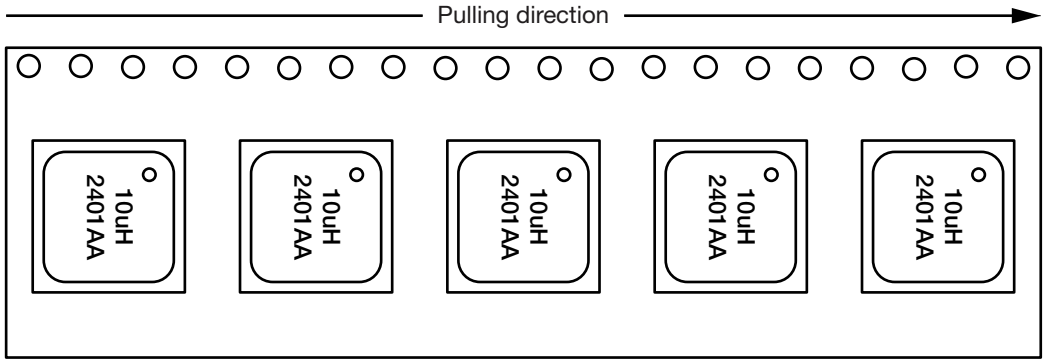
For the IHLP-4040DZ-51 series, the polarity mark also indicates the "start" or "inside" lead of the winding.

Observing the polarity orientation when mounting the inductor will ensure the most consistent EMI reduction performance.

Drawing is for illustrative purposes only. The flux leakage from the inductor is minimal.

PART MARKING / POCKET TAPE ORIENTATION

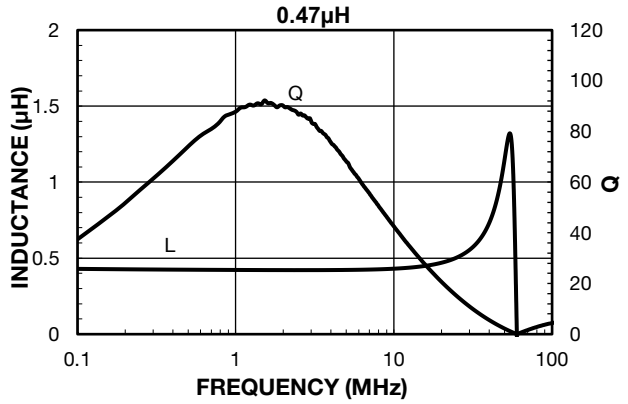
Part marking and tape orientation for IHLP with polarity marking

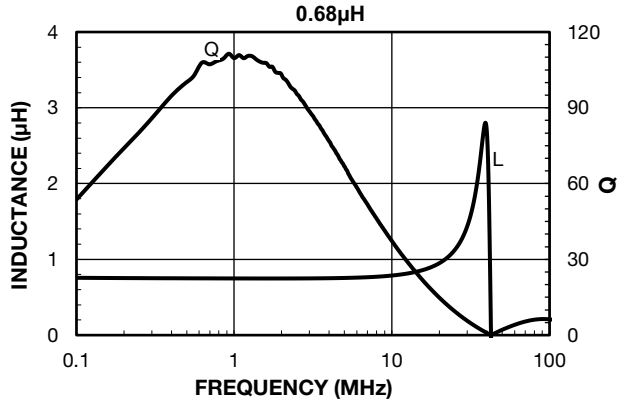


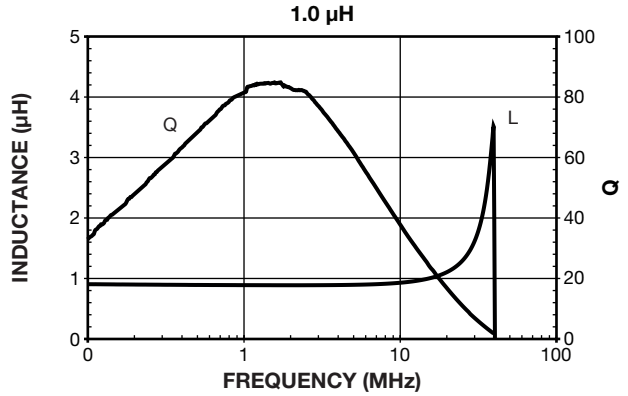
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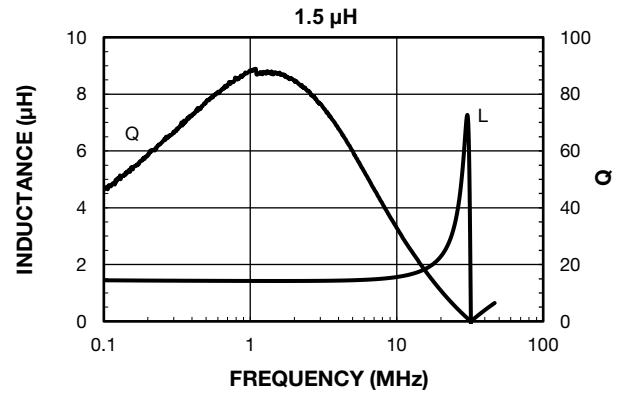
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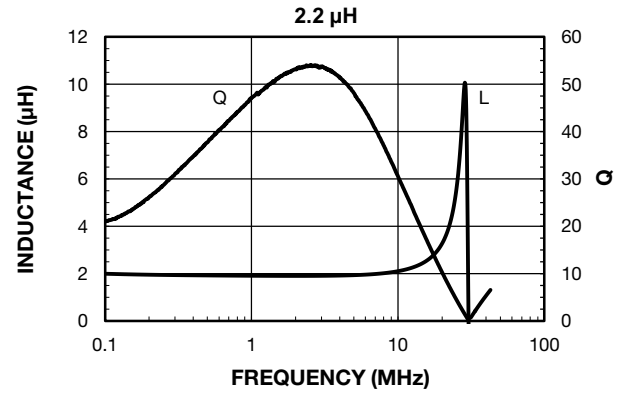
PERFORMANCE GRAPHS

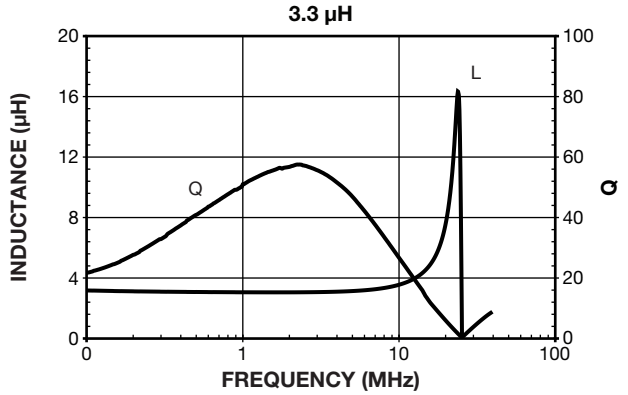


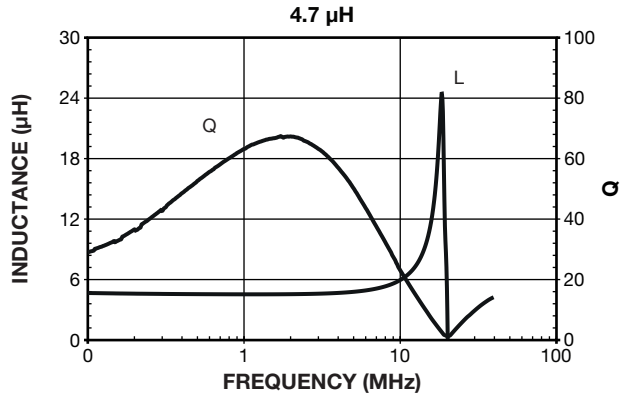


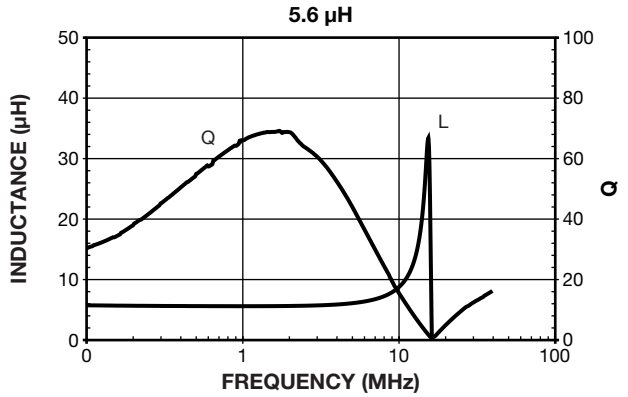








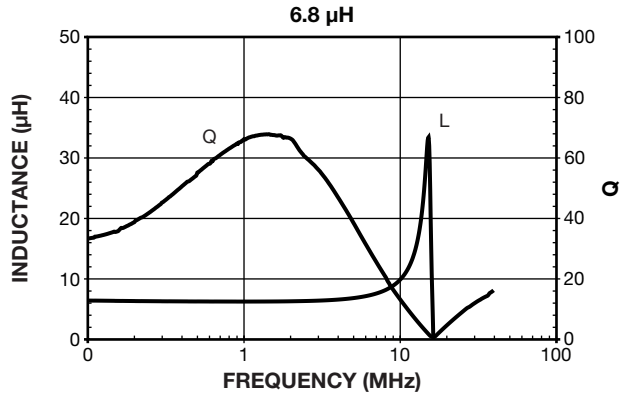


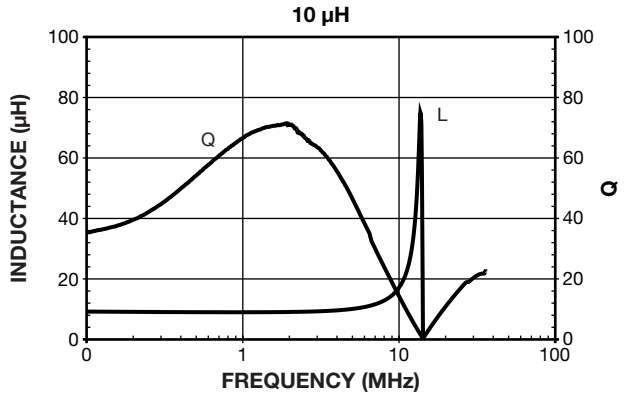


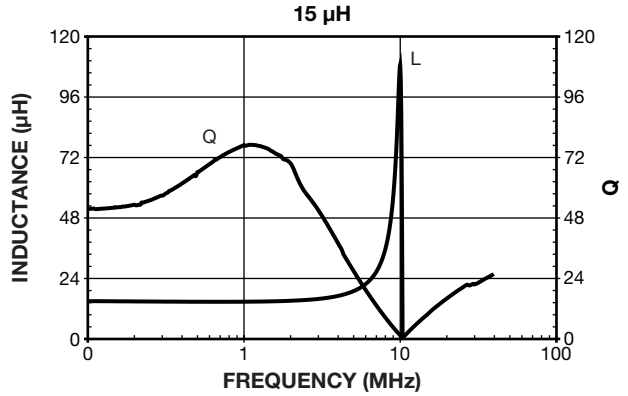
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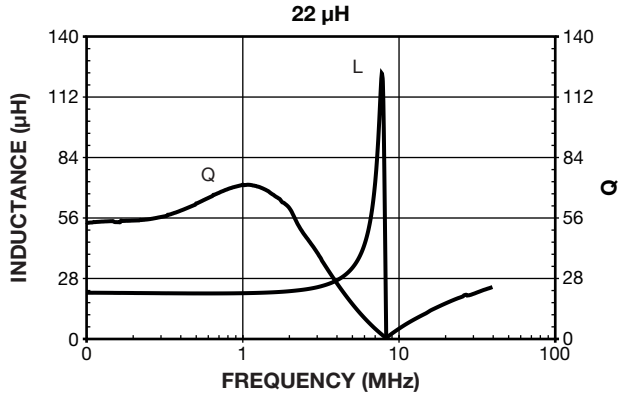
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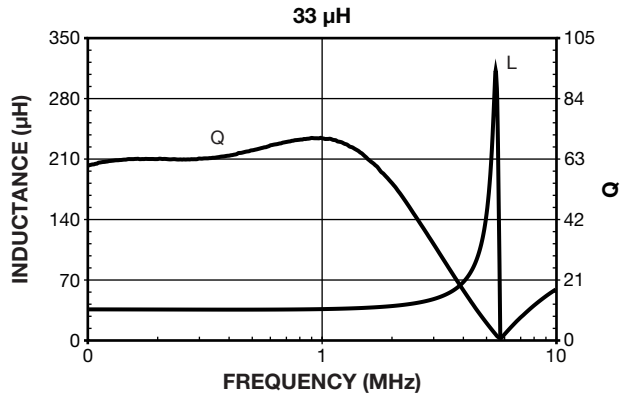
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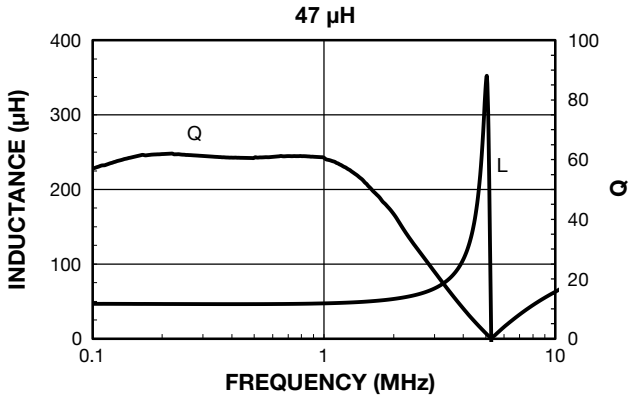


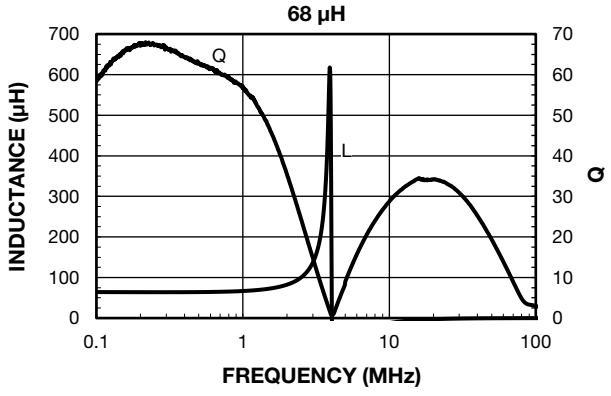








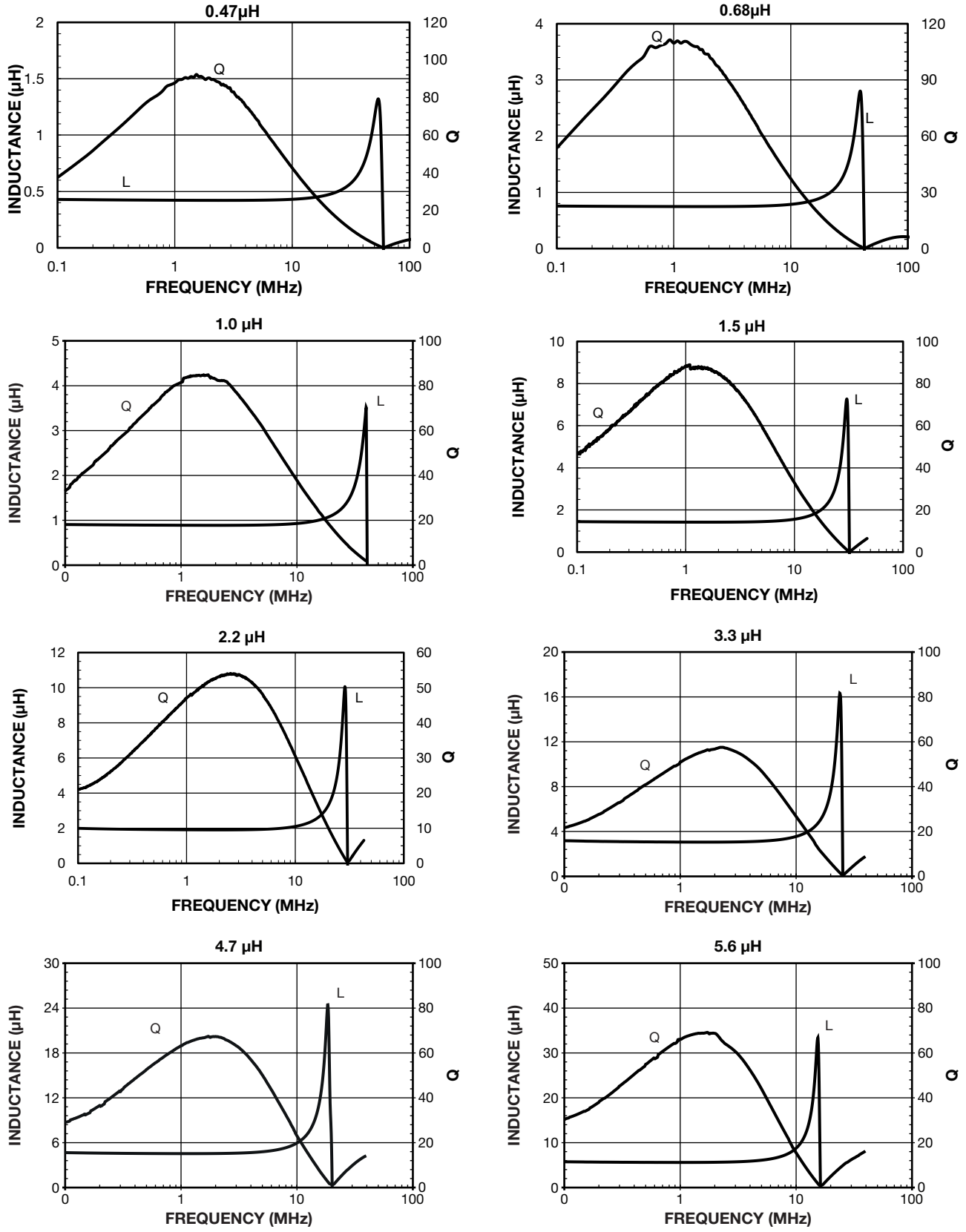




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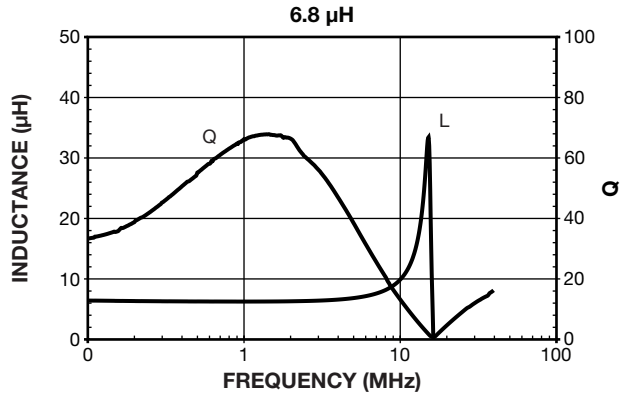
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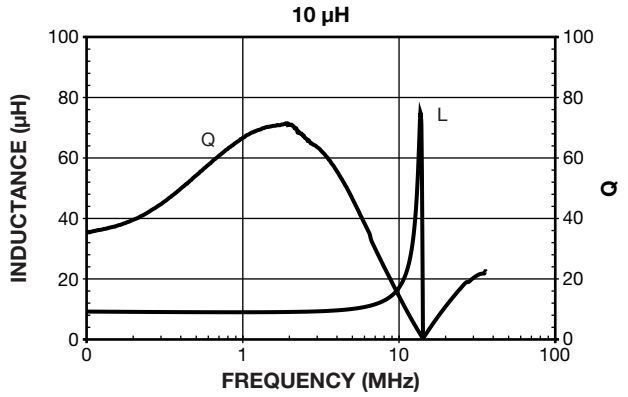
PERFORMANCE GRAPHS: INDUCTANCE AND Q VS. FREQUENCY

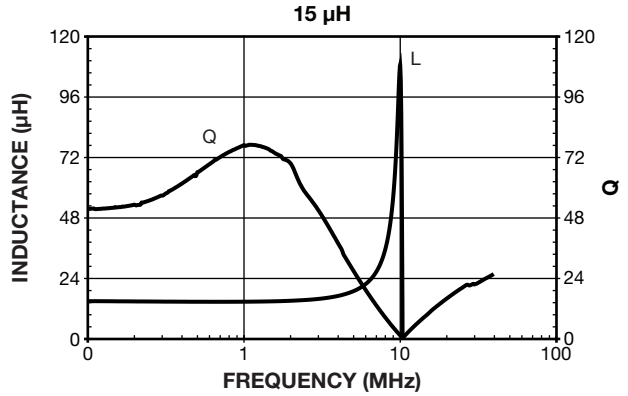


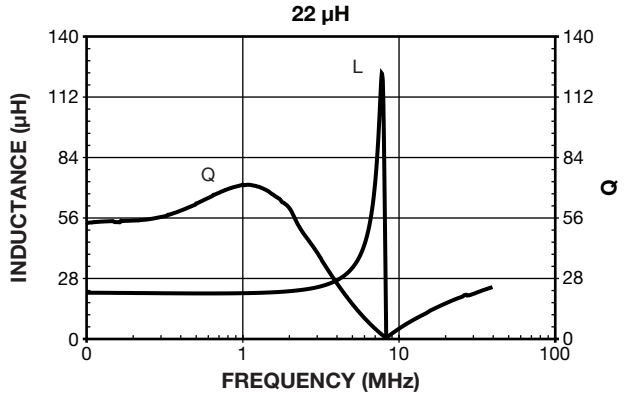
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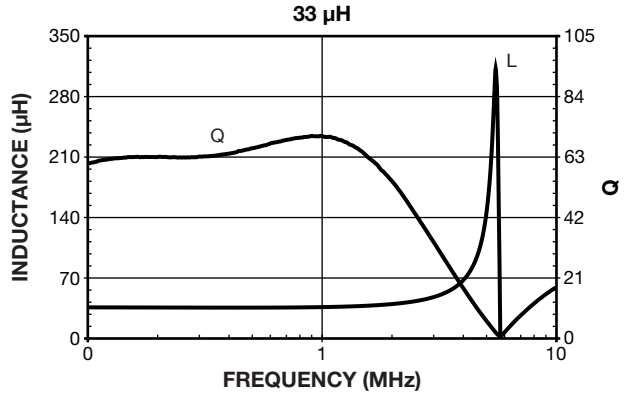
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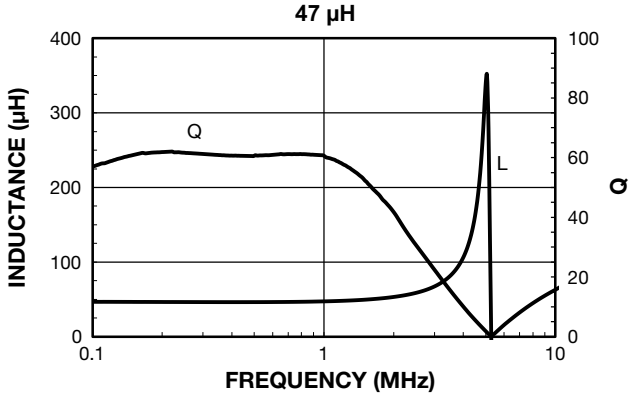


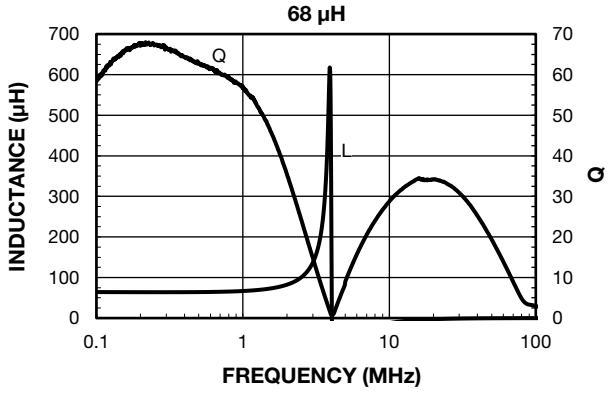












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